T-2: The Branch-Sector Correspondence, Resolved

Two mechanisms, one root. Deliverable per the Season Two plan: the open item from gate note §7, closed by explicit propagator/spectral decomposition, scan-then-report throughout — both mechanisms below survived only after two prior guesses (the nilpotent-tower reading, the same-qualitative-growth reading) died on their own numbers, and this note's own first attempt (a naive per-sector symplectic spectrum, assuming a generic Ω_8) failed the same way before the actual structure was found.

Revision 1.1. The root paragraph's biconditional is split: invariance $\Rightarrow$ nondegeneracy is now stated as the theorem it is; the converse remains per-anchor-verified. Full-band J_0-invariance is upgraded from observed to derived on two of the three classes — a two-line theorem for the pure-antilinear class, an exact cross-Gram equivalence for the mixed extreme class (new, small, previously unstated). The scope section's G3 citation is corrected to its actual boundary (a kernel theorem, not a full-band one). No other content changed.

The root: G3 seen symplectically

The kernel-invariance criterion (inventory G3) — a canonical diagonal anchor's tunnel kernel is J_0-invariant iff L_c, L_a annihilate it jointly — has a symplectic reading that resolves this entire correspondence in two mechanisms rather than one. **Invariance $\Rightarrow$ nondegeneracy is a theorem, not an observation**: for any J_0-invariant subspace and nonzero v in it, $J_0 v$ is also in the subspace, and $\sigma(v, J_0 v) = g(v, v) > 0$ by the compatible-triple relation ($g(x, y) := \sigma(x, J_0 y)$, Foundation's own definition) — σ cannot vanish identically on an invariant subspace, so invariance forces nondegeneracy outright. **The converse — that J_0-transversality forces the Lagrangian property — is not a general symplectic fact; it is verified per-anchor below**, derived exactly for two of the three tested classes. Since full-band J_0-invariance is precisely what separates the 24 extreme-class anchors from the 60 generic ones at the kernel (G3's own scope), the correspondence's mechanism is class-dependent by a criterion this note extends beyond the kernel to the full band structure — confirmed directly on all three integrated anchors, derived rather than merely observed on two of the three classes (below).

Mechanism 1: the generic class pairs across the seam

[FACT, two-implemented] For the expanding anchor ($e_1 + e_{10}$), each of the three T_A-bands is J_0-transverse, and the drift/diffusion-invariant decomposition into band 2 and the coupled kernel+band-4 sector is a genuine Lagrangian pair. Checked directly: Ω restricted to either sector alone is exactly zero (residuals 10^{-17}–10^{-33}), while the cross-block pairing B is invertible ($\det B = -1$). Since $\sigma(t)$ stays block-diagonal under this same decomposition (confirmed: the A, D cross-term between sectors is exactly zero), the symplectic spectrum of the full system is not a property of either block — it is a **geometric mean across the pairing**:

$$\nu(t)^2 = \text{eig}\big(\sigma_2(t) \cdot B \cdot \sigma_c(t) \cdot B^T\big).$$

Tested against the measured branches at $t = 3$: $\{2.3732, 1139.0643\}$ against $\{2.373, 1139.067\}$ — exact to four decimals, no free parameter. This is why both prior stories died: the slow branch (2.373, refusing to be e^3 or e^6) was never one sector's own exponential — it is the paired product of a growing factor and a lingering one, invisible to any reading that looked at a single side of the seam.

Mechanism 2: the extreme classes decompose per band

[THM, two-implemented] Full-band J_0-invariance is derived, not merely observed, on both extreme classes — by two different arguments. For the **pure-antilinear class** ($e_1 + e_{12}$): $\{L_A, J_0\} = 0$ exactly (verified, residual 0.0) — L_A itself anticommutes with J_0. Since J_0 is antisymmetric, L_A^T inherits the same anticommutation, and two anticommutations compose to a commutation: $T_A J_0 = L_A^T L_A J_0 = L_A^T(-J_0 L_A) = -(L_A^T J_0) L_A = -(-J_0 L_A^T) L_A = J_0 T_A$. Every T_A-eigenspace is therefore automatically J_0-invariant — a two-line theorem, confirmed exactly ($\|[T_A, J_0]\| = 0.0$). For the **mixed extreme class** ($e_4 + e_9$): the same anticommutation does not hold for L_A itself ($\|\{L_A, J_0\}\| \approx 9.8$, genuinely nonzero), so $[T_A, J_0] = 0$ is not automatic the same way — but expanding $T_A = L_c^T L_c + L_a^T L_a + L_c^T L_a + L_a^T L_c$ against J_0 shows $[T_A, J_0] = 0$ is exactly equivalent to the cross-Gram condition $L_c^T L_a + L_a^T L_c = 0$, confirmed directly (residual 0.0) — a new, small, exact fact about this class, not previously stated. (For the **generic class**, by contrast, $\|[T_A, J_0]\| = 8.0$ exactly — genuinely nonzero, consistent with the Lagrangian-pair mechanism above rather than per-band invariance.)

[FACT, two-implemented] All three T_A-bands are individually J_0-invariant and Ω-nondegenerate — each is a genuine, self-contained little phase space. Verified directly on the contracting anchor ($e_4 + e_9$): $\det(\Omega \text{ restricted}) = 1.0$ on all three bands. The full correspondence is then simply the restricted symplectic spectrum computed independently on each band, with no pairing needed:

Band	Dimension	Restricted symplectic spectrum at $t = 3$	Identification
Kernel ($T_A = 0$)	4	1.5	The frozen core, exact stasis
$T_A = 2$	8	4.5	The diffusive, linearly-growing branch
$T_A = 4$	4	0.5	The vacuum-landed, decaying branch

All three match the anchor's own measured branches exactly. The three-way character of the contracting anchor's own flow — frozen, growing, decaying — is not three coincidental behaviors; it is three independent phase spaces, one per band, each evolving under its own restricted dynamics.

Scope, honestly

Both mechanisms are verified on one representative anchor of the relevant class (generic: $e_1 + e_{10}$; extreme: $e_4 + e_9$, plus $e_1 + e_{12}$ for the pure-antilinear derivation), not on all 84 by a general argument. G3's own universality (proven for all 84 diagonals) is a **kernel** theorem — T_A-kernel J_0-invariance, joint annihilation by L_c, L_a. This note's extension to the **full band structure** is derived, not merely swept, on two of the three classes: the pure-antilinear class's full-band invariance follows in two lines from $\{L_A, J_0\} = 0$ itself; the mixed extreme class's follows from the cross-Gram condition $L_c^T L_a + L_a^T L_c = 0$, verified exactly on the tested anchor but not yet shown universal across all 12 mixed-class diagonals the way G3's kernel statement is. A full 84-anchor sweep of the cross-Gram condition specifically — the one piece of this note not yet elevated from anchor-verified to class-universal — would be a cheap, mechanical extension if ever wanted; the generic class's own mechanism (the Lagrangian pairing) remains verified on one anchor only, with no derivation offered here for why it should hold generically beyond the single case checked.

What died on the way here, kept on the record

This note's own first attempt at the extreme-class mechanism assumed a generic 8-dimensional block-diagonal Ω for each drift-invariant sector, without checking whether that sector was symplectically closed at all — it wasn't, for the generic anchor, and the resulting spectrum was nonsensical (three values from an eight-dimensional space, no correspondence to anything measured). The error was caught by checking the assumption (is this subspace

Ω-invariant at all?) before trusting its output, the same countermeasure this season named after the stale-J_0-file arc. The corrected reading — that non-isotropy versus isotropy is itself the thing to test, not assumed away — is what produced both mechanisms above.

Compiled by Claude (Anthropic), Program ω Season Two, T-2. All conclusions and any errors remain the author's responsibility.